MIDNIGHT · WAVE 2

datum

Proof of reserves shows the tokens exist. It does not show they can be sold for what the book says.

datum proves, in zero knowledge, that a lending book covers its debt at the price its collateral would actually fetch. The pool is public. The book stays private.

datum.ochinimus.app github.com/seekdaseek/datum Live on Midnight Preprod

THE PROBLEM

Every solvency check values collateral at a price nobody can sell at

WHAT PROOF OF RESERVES SAYS

The tokens exist. Balance-threshold checks such as `proveSolvency` multiply them by the oracle mark and compare.

WHAT IT MISSES

The mark is the last small trade. A lender holding millions of a thin token cannot sell at it: selling moves the pool. The book passes at the mark and fails at the exit.

ON THE SAME PRIVATE BOOK, SAME POOL

10,000,000 AERO against $5,500,000 of debt at a 1.20x bar. A faithful mark-based check in Compact says **COVERED**. datum says **NOT COVERED**. Only the depth differs.

WHO NEEDS IT, WHO PAYS

Lenders whose collateral has one real exit

CeFi LENDERS

Publish proof of reserves today. They can show the reserves cover liabilities at the price they could actually be sold for, without publishing the book.

TOKENIZED STOCK AND RWA LENDERS

Collateral often has one thin on-chain venue, where the mark and the exit diverge most.

DAO TREASURIES

Report runway in their own volatile token, which they could only sell into their own pool.

WHO PAYS, AND HOW MUCH

The lender, or the counterparty who wants the proof: **$250 per published attestation**, or **$1,500 a month** for daily attestations with a hosted verification page. The marginal cost is one proof (seconds of CPU) and one Midnight fee in DUST.

HOW IT WORKS

Multiply and compare. The ledger checks the clock.

PRIVATE (COMPACT WITNESSES)

Position sizes, debt, claimed proceeds per slot, the commitment nonce, the attestor secret, the previous period's opening.

PUBLIC (ON THE LEDGER)

The venue array with its provenance, the required ratio, a one-bit verdict, a hiding commitment chained to the last period, and the time.

```
fee bound   p · (X·10000 + q·g) ≤ Y·q·g    g = 10000 − feeBps
freshness   blockTimeGt(refTime) and blockTimeLt(refTime + maxRefAge)
distinct    no two active slots name the same (chain, pool)
overall     Σp · 1e6 ≥ debt · requiredRatio → covered
```

Each venue slot binds its chain, pool, Sync block, Sync transaction, log index and reference block. The verifier re-fetches that log from Base and checks that no newer Sync exists. One reserve changed by a millionth of a dollar reads MISMATCH.

TWO REAL BOOKS

Moonwell’s WELL and AERO collateral, sold into the pool where it trades

WELL collateral

exit: Aerodrome WELL/WETH, 1% fee

AT THE ORACLE MARK	AT REALISABLE EXIT	DOES NOT EXIST AT EXIT
1.82x	**1.21x**	**$105,064**
$315,634	$210,570 sold into the pool	33.3% of the marked value

131,253,121 WELL supplied to Moonwell against a $173,598 borrowing cap (55% collateral factor, oracle $0.0024048). The WETH leg is valued at Moonwell's own WETH price; through Base's only x·y=k WETH/USDC pool it would fetch 5.1% less.

AERO collateral

exit: Aerodrome AERO/USDC, 0.30% fee

AT THE ORACLE MARK	AT REALISABLE EXIT	DOES NOT EXIST AT EXIT
1.67x	**1.48x**	**$298,134**
$2,647,788	$2,349,653 sold into the pool	11.3% of the marked value

3,404,767 AERO supplied to Moonwell against a $1,588,673 borrowing cap (60% collateral factor, oracle $0.77767). Both books attested at a required ratio of 1.00x in one Preprod run; the debt side is the borrowing cap, not outstanding borrows.

Every input is public on Base: anyone can recompute either book from the page or with `node scripts/books/moonwell.mjs WELL`.

WHAT IS LIVE

On Midnight Preprod, with a hosted prover anyone can use

ON CHAIN (PREPROD)

```
WELL book d2bd781efa…901ac8
attest tx 18c8bd7232…51aecd · block 2,890,448
AERO book 60c0dc62d6…4184f1
attest tx bbf66ac0ff…b297ec · block 2,890,434
demo 763d37058e…e7910c
live attest tx 3088f64a41…5dd89c · block 3,040,865
```

MEASURED ON THE LIVE SERVICE

Attest from the page to verified result: **70 s**.
Proof: **14.7 s**, prover peak **459 MB**.
datum-api at rest: **560 MB**.

Sized for a 2 GB server: the prover is started per job, stopped after five idle minutes, and only started when memory allows, so it cannot take the host down.

ONE LIVE ATTESTATION, STEP BY STEP (seconds)

5	1	14.7	3	16	20	8
read fresh reserves from Base	prepare the private book	prove in zero knowledge	sponsor adds DUST	submit to Midnight	wait for the block	page verifies from chain

datum.ochinimus.app (page) · datum-api.ochinimus.app (attest service, read-only indexer relay). Preprod's indexer is Blockfrost's; the project token stays on the server.

WHAT CHANGED SINCE WAVE 1

From a read-only page to an attestation anyone can make and check

	Wave 1	Wave 2
Venues	Hand-entered reserves	Real Aerodrome pools; chain, pool, Sync block, tx, log index bound into the proof
Provenance	None	Verify from chain; a tampered copy reads MISMATCH
Real books	None	Moonwell WELL and AERO, with WETH quoted at a checked oracle rate
Contract	Last verdict only	Exact fee bound, ledger-enforced freshness, distinct pools, attestor key, chained history
User action	Read-only page	Attest a book from the page; the sponsor pays DUST; the page verifies the result
Tests	22	86 tests, 6 real-ZK checks, 14 of 14 mutants killed
Hosting	Static page	Dedicated small server; Blockfrost indexer behind a read-only relay

JUDGE STEPS

Five minutes, no wallet, no key

1. Open **datum.ochinimus.app**. Press **Recompute from Base** on each book: it shows MATCHES BASE.
2. Press **Verify from chain**: VERIFIED. Then **Tamper with a copy**: MISMATCH on two checks.
3. In **Attest a book yourself**, wait for Sponsor ready, keep the overlevered AERO book and press **Attest this book on Midnight**. Watch the steps to a transaction, a block and NOT COVERED, then the page verifies it itself.

```
git clone https://github.com/seekdaseek/datum && cd datum
npm install && npm run build && npm test   # 86 pass
BLOCKFROST_PROJECT_ID=… node scripts/verify.mjs --network preprod
node scripts/verify.mjs --network preprod --tamper   # MISMATCH, exit 1
```

HONEST LIMITS

Where this method is the wrong tool

Concentrated liquidity (Uniswap v3, Aerodrome Slipstream): depth depends on tick ranges. MORPHO on Base exits mainly there, so datum does not price it.

Aggregators and routers: a route is not one reserve pair.

Depth quoted in another volatile asset: WELL's depth is in WELL/WETH and is priced through Moonwell's WETH rate, not a second hop.

Off-chain exits and stable-swap pools: nothing to re-fetch, or a different invariant.

The attestor declares the book. The attestor key makes each declaration accountable and chained to the last; for the two Moonwell books every input is public, so anyone recomputes them. Where datum applies it is a lower bound.

NEXT STEPS AND BUSINESS DEVELOPMENT

Wave 3: price the boundary, then put it in front of lenders

PRODUCT

A second hop for depth quoted in a volatile asset. A conservative bound for concentrated liquidity from tick data. A one-command lender-side prover. Scheduled attestations of a real book. A Midnight contract that consumes another contract's verdict.

ADOPTION

1. Open source today: the prover, verifier and page run against any Aerodrome volatile pool.
2. Hosted attestation for a lender: their prover runs locally, we host the page and schedule it.
3. Verdicts consumed by risk dashboards, curators and counterparties reading the indexer.

First conversations: lenders that already publish proof of reserves, and RWA lenders with one thin venue. Mainnet when Midnight mainnet accepts it.